Homework 4

Alejandro De La Torre
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Instructor: Thierry Laurens

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Problem 1

Let $\{a_n\}_{n\geq 1}$ be a sequence.

(a) Prove that if $\{a_n\}_{n\geq 1}$ converges to a, then the sequence $\{|a_n|\}_{n\geq 1}$ converges to $|a|$. (Hint: $||x| - |y|| \leq |x - y|$ for any $x, y \in \mathbb{R}$.)

(b) Provide an example where $\{|a_n|\}_{n\geq 1}$ converges, but $\{a_n\}_{n\geq 1}$ diverges.

Discussion & Scratch Work

- **(a)** The idea is: if $a_n \to a$, then eventually a_n is very close to a. The absolute value function is "distance from 0," so if two numbers are close, their distances from 0 are also close. The key inequality is

$$\big||x| - |y|\big| \leq |x - y|,$$

  which lets us bound $\big||a_n| - |a|\big|$ by $|a_n - a|$ and then use the definition of convergence.
- **(b)** To make $|a_n|$ converge while a_n diverges, we can alternate signs: take $a_n = (-1)^n$. Then $|a_n| = 1$ for all n (so it converges), but a_n oscillates between 1 and -1 (so it cannot converge).

Proof.
(a) Assume $a_n \to a$. Let $\varepsilon > 0$ be given. By convergence, there exists $N \in \mathbb{N}$ such that for all $n \geq N$ we have $|a_n - a| < \varepsilon$. Using the inequality $\big||x| - |y|\big| \leq |x - y|$ (valid for all $x, y \in \mathbb{R}$), with $x = a_n$ and $y = a$, we obtain for all $n \geq N$:

$$\big||a_n| - |a|\big| \leq |a_n - a| < \varepsilon.$$

Hence, by the definition of convergence, $|a_n|$ converges to $|a|$.

(b) Define $a_n := (-1)^n$. Then $|a_n| = 1$ for all n, so $\{|a_n|\}$ converges to 1. However, $\{a_n\}$ does not converge: if it converged to some $L \in \mathbb{R}$, then the subsequences $a_{2k} = 1$ and $a_{2k+1} = -1$ would both have to converge to L, forcing $1 = L = -1$, a contradiction. Therefore $\{|a_n|\}$ converges while $\{a_n\}$ diverges. $\square$

Problem 2

(Squeeze theorem). Prove that if $\{a_n\}_{n\geq 1}$, $\{b_n\}_{n\geq 1}$, and $\{c_n\}_{n\geq 1}$ are sequences such that $a_n \leq b_n \leq c_n$ for all $n \geq 1$ and $\lim_{n\to\infty} a_n = L = \lim_{n\to\infty} c_n$, then $\{b_n\}_{n\geq 1}$ converges and $\lim_{n\to\infty} b_n = L$.

Discussion & Scratch Work

- Intuition: b_n is "trapped" between a_n and c_n. If both outer sequences converge to the same limit L, then for any small tolerance $\varepsilon > 0$, eventually both a_n and c_n lie inside the interval $(L - \varepsilon,\, L + \varepsilon)$. Since $a_n \leq b_n \leq c_n$, the middle term must also lie in that same interval, forcing $b_n \to L$.
- We will use the definition of convergence:

  $$a_n \to L \iff \forall \varepsilon > 0\ \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N,\ |a_n - L| < \varepsilon,$$

  (and similarly for c_n). The only "construction" step is choosing a single index after which *both* conditions hold; we do this by taking a maximum.

Proof.
Assume that $a_n \leq b_n \leq c_n$ for all $n \geq 1$, and that $\lim_{n\to\infty} a_n = L = \lim_{n\to\infty} c_n$. We prove that $b_n \to L$.

Let $\varepsilon > 0$ be given. Since $a_n \to L$, there exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$,

$$|a_n - L| < \varepsilon \quad \Longrightarrow \quad L - \varepsilon < a_n < L + \varepsilon.$$

Similarly, since $c_n \to L$, there exists $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$,

$$|c_n - L| < \varepsilon \quad \Longrightarrow \quad L - \varepsilon < c_n < L + \varepsilon.$$

Set $N := \max\{N_1, N_2\}$. Then for all $n \geq N$, both inequalities hold, and we have

$$L - \varepsilon < a_n \leq b_n \leq c_n < L + \varepsilon.$$

In particular, for all $n \geq N$,

$$L - \varepsilon < b_n < L + \varepsilon,$$

which is equivalent to $|b_n - L| < \varepsilon$. Since $\varepsilon > 0$ was arbitrary, this proves that $b_n \to L$. $\square$

Problem 3

Suppose that the sequence $\{a_n\}_{n\geq 1}$ converges.

(a) Prove that if $a_n \geq m$ for all but finitely many n, then $\lim_{n\to\infty} a_n \geq m$.

(b) Prove that if $a_n \leq M$ for all but finitely many n, then $\lim_{n\to\infty} a_n \leq M$.

Discussion & Scratch Work

- The key idea is: convergence prevents the sequence from staying *strictly below* a number that it is eventually always above. If $a_n \to \ell$ and (eventually) $a_n \geq m$, then the limit *cannot* satisfy $\ell < m$, because choosing $\varepsilon = \frac{m-\ell}{2}$ would force $a_n < \ell+\varepsilon < m$ for all large n, contradicting $a_n \geq m$ for all large n.
- Part (b) is the symmetric argument: if (eventually) $a_n \leq M$, then the limit cannot be $> M$.

Proof.
Let $\ell := \lim_{n\to\infty} a_n$.

(a) Assume that $a_n \geq m$ for all but finitely many n. Then there exists $N_0 \in \mathbb{N}$ such that

$$a_n \geq m \quad \text{for all } n \geq N_0.$$

We claim that $\ell \geq m$. Suppose for contradiction that $\ell < m$. Set

$$\varepsilon := \frac{m-\ell}{2} > 0.$$

Since $a_n \to \ell$, there exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$,

$$|a_n - \ell| < \varepsilon \quad \Longrightarrow \quad a_n < \ell + \varepsilon = \frac{\ell+m}{2} < m.$$

Let $N := \max\{N_0, N_1\}$. Then for all $n \geq N$ we have simultaneously $a_n \geq m$ and $a_n < m$, a contradiction. Hence $\ell \geq m$.

(b) Assume that $a_n \leq M$ for all but finitely many n. Then there exists $N_0 \in \mathbb{N}$ such that $a_n \leq M$ for all $n \geq N_0$. We claim that $\ell \leq M$. Suppose for contradiction that $\ell > M$. Set $\varepsilon := \frac{\ell-M}{2} > 0$. Since $a_n \to \ell$, there exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$,

$$|a_n - \ell| < \varepsilon \quad \Longrightarrow \quad a_n > \ell - \varepsilon = \frac{\ell+M}{2} > M.$$

Let $N := \max\{N_0, N_1\}$. Then for all $n \geq N$ we have simultaneously $a_n \leq M$ and $a_n > M$, a contradiction. Hence $\ell \leq M$. $\square$

Problem 4

Suppose that the sequence $\{a_n\}_{n\geq 1}$ converges. Prove or disprove the following statements:

(a) If $\lim_{n\to\infty} a_n \geq m$, then there exists $N \in \mathbb{N}$ so that $a_n \geq m$ for all $n \geq N$.

(b) If $\lim_{n\to\infty} a_n > m$, then there exists $N \in \mathbb{N}$ so that $a_n > m$ for all $n \geq N$.

Discussion & Scratch Work

- **Key idea.** Part (a) smells suspicious because the hypothesis allows equality: $\lim a_n \geq m$ includes the borderline case $\lim a_n = m$. Convergence to m does *not* force the tail of the sequence to lie on one side of m; it only forces the tail to lie *near* m. So we should try to build a convergent sequence with limit m that stays strictly below m forever.
- Part (b) replaces $\geq$ by $>$, which creates a positive gap $L-m$. If $a_n \to L$ and $L > m$, then choosing ε smaller than this gap forces a_n to eventually lie in $(L-\varepsilon, L+\varepsilon)$, which sits entirely above m. So (b) should be true, and the proof will be a direct ε–N argument.

Proof.
Let $\ell := \lim_{n\to\infty} a_n$.

(a) Disprove. The statement is false. Fix $m \in \mathbb{R}$ and define

$$a_n := m - \frac{1}{n} \qquad (n \geq 1).$$

Then $a_n \to m$ as $n \to \infty$, so $\lim_{n\to\infty} a_n = m \geq m$ holds. However, for every $n \geq 1$ we have $a_n = m - \frac{1}{n} < m$, so there is *no* $N \in \mathbb{N}$ such that $a_n \geq m$ for all $n \geq N$. Hence (a) is false.

(b) Prove. Assume $\ell > m$. Set

$$\varepsilon := \frac{\ell - m}{2} > 0.$$

Since $a_n \to \ell$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$|a_n - \ell| < \varepsilon \quad \Longrightarrow \quad a_n > \ell - \varepsilon = \ell - \frac{\ell - m}{2} = \frac{\ell + m}{2} > m.$$

Therefore $a_n > m$ for all $n \geq N$, as desired. $\square$

Problem 5

Prove that if $\{a_n\}_{n\geq 1}$ is a sequence of positive numbers that converges to $a > 0$, then the sequence $\{\sqrt{a_n}\}_{n\geq 1}$ converges to $\sqrt{a}$.

Discussion & Scratch Work

- The main trick is to rewrite the difference of square-roots by "rationalizing":

$$|\sqrt{x} - \sqrt{y}| = \frac{|x-y|}{\sqrt{x}+\sqrt{y}} \qquad (x, y \geq 0).$$

  Here we will take $x = a_n$ and $y = a$.
- Because $a > 0$, the denominator $\sqrt{a_n} + \sqrt{a}$ is always at least $\sqrt{a}$ (since $a_n > 0 \Rightarrow \sqrt{a_n} \geq 0$). So we can control $|\sqrt{a_n} - \sqrt{a}|$ directly by controlling $|a_n - a|$.
- Concretely, given $\varepsilon > 0$, it suffices to make $|a_n - a| < \varepsilon\sqrt{a}$.

Proof.
Let $\varepsilon > 0$ be given. Since $a_n \to a$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$|a_n - a| < \varepsilon\sqrt{a}.$$

For $n \geq N$, we have $a_n > 0$ and $a > 0$, so $\sqrt{a_n}$ and $\sqrt{a}$ are defined and nonnegative. Using

$$(\sqrt{a_n} - \sqrt{a})(\sqrt{a_n} + \sqrt{a}) = a_n - a,$$

we obtain

$$|\sqrt{a_n} - \sqrt{a}| = \frac{|a_n - a|}{\sqrt{a_n} + \sqrt{a}} \leq \frac{|a_n - a|}{\sqrt{a}}.$$

Therefore, for all $n \geq N$,

$$|\sqrt{a_n} - \sqrt{a}| \leq \frac{|a_n - a|}{\sqrt{a}} < \frac{\varepsilon\sqrt{a}}{\sqrt{a}} = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, it follows that $\sqrt{a_n} \to \sqrt{a}$. $\square$

Problem 6

Consider the following sequence: $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2 + a_n}$ for all $n \geq 1$.

(a) Prove that the sequence $\{a_n\}_{n\geq 1}$ is bounded.

(b) Prove that the sequence is increasing.

(c) Conclude that the sequence converges, and find its limit.

Discussion & Scratch Work

- **What we need.** To show the sequence is *bounded*, we need to find real numbers m, M such that $m \leq a_n \leq M$ for all $n \geq 1$ (equivalently: bounded below and bounded above). To show it is *increasing*, we need $a_n \leq a_{n+1}$ for all $n \geq 1$.
- **Boundedness idea.** Because $a_{n+1} = \sqrt{2 + a_n}$, a natural guess is that the sequence stays below 2: if $a_n \leq 2$, then $a_{n+1} = \sqrt{2 + a_n} \leq \sqrt{4} = 2$. Also $a_n \geq 0$ implies $a_{n+1} \geq 0$. Since $a_1 = \sqrt{2} \in (0, 2)$, induction should give $0 < a_n \leq 2$ for all n.
- **Monotonicity idea.** Since all terms are positive, we can compare a_{n+1} and a_n by squaring:
$$a_{n+1} \geq a_n \iff \sqrt{2 + a_n} \geq a_n \iff 2 + a_n \geq a_n^2.$$
The last inequality is $(a_n - 2)(a_n + 1) \leq 0$, which is true whenever $-1 \leq a_n \leq 2$. Our boundedness work gives $0 < a_n \leq 2$, so this will force $a_{n+1} \geq a_n$.
- **Convergence + limit.** Once we know $\{a_n\}$ is increasing and bounded, we may invoke the Monotone Convergence Theorem (proved in lecture): the sequence converges. If $a_n \to L$, then passing to the limit in the recursion gives $L = \sqrt{2 + L}$, so $L^2 - L - 2 = 0$ and $L \in \{2, -1\}$. Since $L \geq 0$, we conclude $L = 2$.

Proof.
(a) Boundedness. We first show that $0 < a_n \leq 2$ for all $n \geq 1$.

Base case. We have $a_1 = \sqrt{2}$, hence $0 < a_1 \leq 2$.

Inductive step. Assume $0 < a_n \leq 2$ for some $n \geq 1$. Then $2 + a_n \leq 4$, so
$$a_{n+1} = \sqrt{2 + a_n} \leq \sqrt{4} = 2.$$
Also $2 + a_n > 0$, so $a_{n+1} = \sqrt{2 + a_n} > 0$. Thus $0 < a_{n+1} \leq 2$.

By induction, $0 < a_n \leq 2$ for all $n \geq 1$. In particular, the sequence is bounded (below by 0 and above by 2).

(b) Increasing. Fix $n \geq 1$. From part (a), $a_n > 0$, so we may square inequalities without changing direction. We compute:
$$a_{n+1} \geq a_n \iff \sqrt{2 + a_n} \geq a_n \iff 2 + a_n \geq a_n^2 \iff a_n^2 - a_n - 2 \leq 0 \iff (a_n - 2)(a_n + 1) \leq 0.$$
But part (a) gives $0 < a_n \leq 2$, hence $a_n - 2 \leq 0$ and $a_n + 1 > 0$, so indeed $(a_n - 2)(a_n + 1) \leq 0$. Therefore $a_{n+1} \geq a_n$ for all $n \geq 1$, and the sequence is increasing.

(c) Convergence and limit. By parts (a) and (b), $\{a_n\}$ is bounded and increasing, so by the Monotone Convergence Theorem it converges. Let

$$L := \lim_{n\to\infty} a_n.$$

Because $a_{n+1} = \sqrt{2 + a_n}$ and the square-root function is continuous on $(0, \infty)$, we may take limits on both sides to obtain

$$L = \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \sqrt{2 + a_n} = \sqrt{2 + \lim_{n\to\infty} a_n} = \sqrt{2 + L}.$$

Squaring gives $L^2 = 2 + L$, i.e.

$$L^2 - L - 2 = 0 \quad \Longrightarrow \quad (L - 2)(L + 1) = 0.$$

Thus $L \in \{2, -1\}$. Since each $a_n > 0$, we have $L \geq 0$, so $L \neq -1$ and therefore $L = 2$. $\square$

Appendix: Toolbox

Completeness / Real numbers

Proposition 1 (Archimedean property)**.** *For every $x \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $n > x$.*

Corollary 1. *For every $\varepsilon > 0$, there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$.*

Lemma (Integer "sandwich")**.** *For every $a \in \mathbb{R}$, there exists $N \in \mathbb{Z}$ such that*

$$N \leq a < N + 1.$$

Proposition 2 ($\mathbb{Q}$ is dense in $\mathbb{R}$)**.** *If $x, y \in \mathbb{R}$ satisfy $x < y$, then there exists $q \in \mathbb{Q}$ such that*

$$x < q < y.$$

Corollary 2 ($\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$)**.** *If $x, y \in \mathbb{R}$ satisfy $x < y$, then there exists $r \in \mathbb{R} \setminus \mathbb{Q}$ such that*

$$x < r < y.$$

Countability

Proposition 3 (Uncountability via Cantor diagonal)**.** *Let*

$$A := \{f : \mathbb{N} \to \{0, 1\}\}.$$

Then A is uncountable.

Remark (Cantor's diagonal argument (what is happening conceptually))**.** *If we (incorrectly) assume A is countable, we can list its elements as*

$$A = \{f_1, f_2, f_3, \dots\}, \qquad f_k : \mathbb{N} \to \{0, 1\}.$$

Think of this as an infinite table whose (k, n)-entry is $f_k(n) \in \{0, 1\}$. We then define a new function

$$g : \mathbb{N} \to \{0, 1\}, \qquad g(n) := 1 - f_n(n).$$

This flips the diagonal entries. For each $n \in \mathbb{N}$ we have

$$g(n) \neq f_n(n),$$

so $g \neq f_n$ for every n (they differ at input n). Thus $g \in A$ but g is not in our purported enumeration, contradicting that the list contained all of A.

Corollary 3. *The interval $(0, 1) = \{x \in \mathbb{R} : 0 < x < 1\}$ is uncountable.*

Sequences and limits

Definition 1 (Sequence)**.** *A* sequence of real numbers *is a function* $f : \mathbb{N} \to \mathbb{R}$*. We write* $a_n := f(n)$ *and denote the sequence by*

$$(a_n)_{n\geq 1} \quad \textit{or} \quad \{a_n\}_{n\geq 1}.$$

Definition 2 (Absolute value)**.** *For* $x \in \mathbb{R}$*,*

$$|x| = \begin{cases} x & \textit{if } x \geq 0, \\ -x & \textit{if } x < 0. \end{cases}$$

Exercise 1 (Basic properties of $|\cdot|$)**.** *For all* $x, y \in \mathbb{R}$*:*

(a) $|x| \geq 0$.

(b) $|x| = 0 \iff x = 0$.

(c) $|x + y| \leq |x| + |y|$ *(triangle inequality).*

(d) $\big||x| - |y|\big| \leq |x - y|$.

(e) $|xy| = |x||y|$.

Definition 3 (Limit / convergence)**.** *We say that the sequence* $\{a_n\}_{n\geq 1}$ converges *to* $a \in \mathbb{R}$ *if*

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N} \textit{ s.t. } \forall n \geq N,\ |a_n - a| < \varepsilon.$$

In this case we write $\lim_{n\to\infty} a_n = a$ *or* $a_n \to a$ *as* $n \to \infty$*.*

Definition 4 (Divergence)**.** *A sequence* diverges *if it does not converge.*

Lemma (Uniqueness of limits)**.** *If* $\{a_n\}_{n\geq 1}$ *converges, then its limit is unique.*

Lemma (Convergent $\Rightarrow$ bounded)**.** *If* $\{a_n\}_{n\geq 1}$ *converges, then it is bounded.*

Definition 5 (Bounded sequence)**.** *We say* $\{a_n\}_{n\geq 1}$ *is bounded above / bounded below / bounded if the set* $\{a_n : n \geq 1\}$ *is bounded above / bounded below / bounded in* $\mathbb{R}$*.*

Theorem 1 (Scalar multiple)**.** *If* $a_n \to a$ *in* $\mathbb{R}$ *and* $k \in \mathbb{R}$*, then* $(ka_n) \to ka$*.*

Theorem 2 (Limit laws)**.** *If* $a_n \to a$ *and* $b_n \to b$*, then:*

(a) $a_n + b_n \to a + b$.

(b) $a_n b_n \to ab$.

(c) *If* $b \neq 0$ *and* $b_n \neq 0$ *for all* n*, then* $\dfrac{a_n}{b_n} \to \dfrac{a}{b}$.

Definition 6 (Monotone sequences)**.** *A sequence* $\{a_n\}_{n\geq 1}$ *is*

(a) increasing *if* $a_n \leq a_{n+1}$ *for all* $n \geq 1$,

(b) strictly increasing *if* $a_n < a_{n+1}$ *for all* $n \geq 1$,

(c) decreasing *if* $a_n \geq a_{n+1}$ *for all* $n \geq 1$,

(d) strictly decreasing *if* $a_n > a_{n+1}$ *for all* $n \geq 1$,

(e) monotone *if it is increasing or decreasing.*

Theorem 3 (Monotone convergence theorem)**.** *If* $\{a_n\}_{n\geq 1}$ *is monotone and bounded, then it converges.*

Cauchy sequences and subsequences

Definition 7 (Cauchy sequence)**.** *A sequence* $\{a_n\}_{n\geq 1}$ *is* Cauchy *if*

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \text{ s.t. } \forall n, m \geq N, \ |a_n - a_m| < \varepsilon.$$

Theorem 4 (Cauchy criterion)**.** *A sequence of real numbers is Cauchy if and only if it converges.*

Lemma (Cauchy $\Rightarrow$ bounded)**.** *If* $\{a_n\}_{n\geq 1}$ *is Cauchy, then it is bounded.*

Definition 8 (Subsequence)**.** *Let* $\{a_n\}_{n\geq 1}$ *be a sequence. We say* $\{a_{h_n}\}_{n\geq 1}$ *is a* subsequence *of* $\{a_n\}_{n\geq 1}$ *if* $\{h_n\}_{n\geq 1}$ *is a strictly increasing sequence of natural numbers (i.e.* $h_n \in \mathbb{N}$ *and* $h_n < h_{n+1}$ *for all* n*).*

Lemma. *If* $\{h_n\}_{n\geq 1}$ *is strictly increasing in* $\mathbb{N}$*, then* $h_n \geq n$ *for all* $n \geq 1$.

Example 1 (Subsequences)**.** ***(a)*** *If* $h_n = n$*, then* $\{a_{h_n}\}$ *is the original sequence.*

(b) *If* $h_n = 2n$*, then* $\{a_{2n}\}$ *is the even-index subsequence.*

(c) *If* $a_n = (-1)^n$*, then the subsequence* $a_{2n} = 1$ *converges (even though* a_n *diverges).*

Theorem 5 (Subsequence of a convergent sequence)**.** *If* $a_n \to a$ *and* $\{a_{h_n}\}$ *is any subsequence, then* $a_{h_n} \to a$.

Theorem 6 (Convergence detected by subsequences)**.** *A sequence* $\{a_n\}$ *converges to* a *if and only if every subsequence of* $\{a_n\}$ *converges to* a.